LEARNER GUIDE

Because–therefore: quadrilaterals and missing angles

Angles Tell Us Why · Lesson 4. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

A **quadrilateral** has four sides and four angles. One **diagonal** splits it into two triangles, so its angles add up to **180° + 180° = 360°**. Special quadrilaterals have extra facts: a **parallelogram**'s opposite angles are equal; a **rectangle**'s angles are all right angles; a **kite** has one pair of equal opposite angles. When one fact leads to the next, write each step with its reason: **because** … **therefore** …

What you are learning to notice

- The angles of every quadrilateral add up to 360°, because a diagonal makes two triangles.
- Three angles of a quadrilateral always decide the fourth.
- Parallelogram: opposite angles equal. Rectangle: four right angles. Kite: one pair of equal opposite angles.
- A chain of reasoning is a list of steps, each with its own reason.
- A measurement can suggest an answer; only reasons prove it.

Moving through it, one step at a time

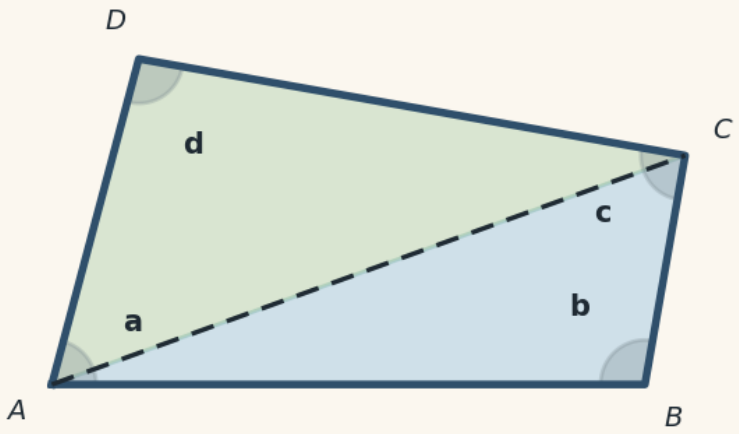
Look at the diagram and decide which fact gives you the next angle. Write the step and its reason on one line, then use the new angle in the next step. Keep going until you reach the angle you were asked for. Check: do the four angles of each quadrilateral add up to 360°?

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Why 360°?

Why do the angles a, b, c and d add up to 360°?

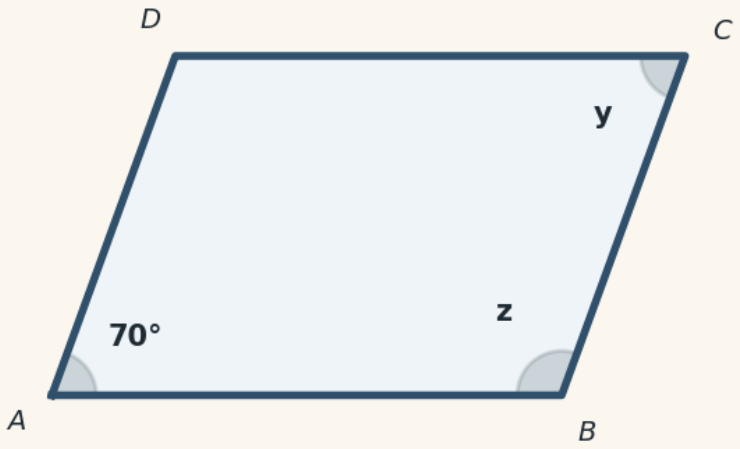


1. Draw the diagonal AC.	two triangles: ABC and ACD
2. The angles of each triangle add up to	180°
3. Together they fill the four corners exactly.	180° + 180° = 360°

a + b + c + d = 360°, for every quadrilateral.

Example 2 — A chain in a parallelogram

ABCD is a parallelogram. Find y, then z.



1. y is opposite the 70° angle.	y = 70°
2. because	opposite angles of a parallelogram are equal
3. 70 + 70 + z + z = 360, so z + z = 220.	z = 110°
4. because	angles in a quadrilateral add up to 360°

y = 70° and z = 110°: two steps, each with its reason.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (angles on a straight line (Lesson 2) and in a triangle (Lesson 3)), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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